

2018 RI Prelim Exam Paper 1

- 1** Adult tickets for a parade are sold at three different prices, depending on the type of seats.

Children under the age of 12 and Senior Citizens aged 65 and above can enjoy 20% and 10% off the adult ticket prices respectively. Those who are between 12 and 64 years old (inclusive) will have to pay the full adult ticket price.

The number of tickets sold in each category for each group of people, together with the total cost of the tickets for each group, are given in the following table.

	Category 1 tickets (premier)	Category 2 tickets (sheltered)	Category 3 tickets (unsheltered)	Total cost
Adults (12 to 64 years old (inclusive))	150	120	200	\$18 000
Children (under 12 years old)	80	40	100	\$6 760
Senior Citizens (65 years old and above)	100	100	100	\$10 665

Write down and solve equations to find the price of an adult ticket for each of the ticket categories. [4]

1 [4]	Let the cost of an adult ticket for Category 1, 2 and 3 be \$$x$, \$$y$ and \$$z$ respectively. $150x + 120y + 200z = 18000$ $80(0.8x) + 40(0.8y) + 100(0.8z) = 6760$ $100(0.9x) + 100(0.9y) + 100(0.9z) = 10665$ $x = 50, \quad y = 40, \quad z = 28.5$ The cost of a Category 1, 2 and 3 adult ticket is \$50, \$40 and \$28.50 respectively.
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- 2 Aden, a member of Raffles Art Club, is helping to design a graphic for the school's homecoming event. The main body of the graphic consists of N concentric circles $C_1, C_2, C_3, \dots, C_N$ with radii $r, 2r, 3r, \dots, Nr$, where r is a constant and N is an even integer, as shown in Fig. 1.

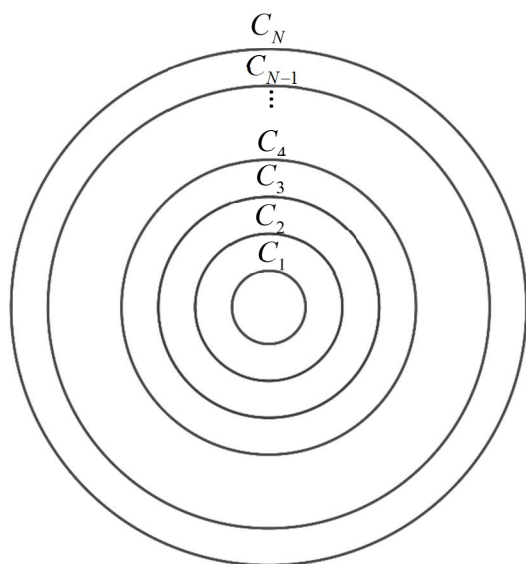


Fig. 1

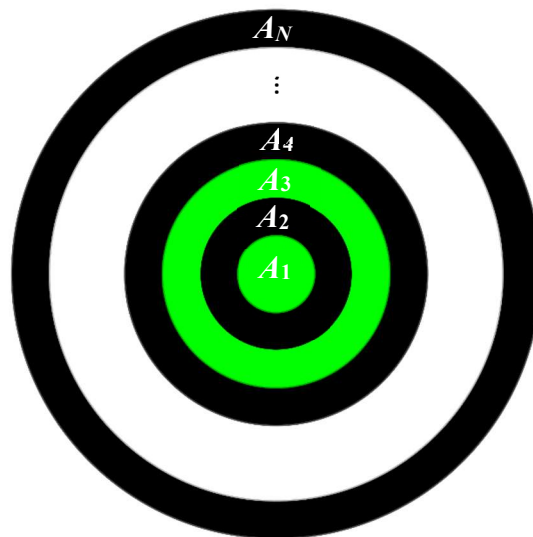


Fig. 2

The region enclosed by the circle C_1 is denoted by A_1 with area a_1 and the region between the circles C_{n-1} and C_n is denoted by A_n with area a_n for $n = 2, 3, 4, \dots, N$, as shown in Fig. 2.

- (i) Show that the sequence $a_1, a_2, a_3, \dots, a_N$ is an arithmetic progression. [4]

With the help of a graphic software, Aden fills $A_1, A_2, A_3, \dots, A_N$ with two of the school colours: green and black. A_n will be filled green if n is odd and will be filled black if n is even.

Aden wishes to create a better visual effect by having different intensities of green. He fills A_1 with the green colour that is the same as the school colour. He reduces the intensity of this green colour by ten percent to fill A_3 ; and reduces the intensity of the green colour used to fill A_3 by ten percent to fill A_5 ; and this process continues for all odd values of n . When he finishes filling up all the areas, Aden finds that the intensity of the green colour that fills A_{N-1} falls below one quarter of the intensity of the school colour for the first time.

- (ii) Find the value of N . [4]

2(i) [4]	$a_n = \pi(nr)^2 - \pi[(n-1)r]^2 \quad \text{for } n = 2, 3, \dots, N$ $= \pi[nr - (n-1)r][nr + (n-1)r]$ $= \pi(r)(2nr - r)$ $= \pi r^2(2n - 1)$ $a_1 = \pi r^2 = \pi r^2(2 \times 1 - 1) \text{ and it follows the form } a_n = \pi r^2(2n - 1) \text{ when } n = 1.$	Read word problems carefully and clarify understanding of the symbols used. Candidates mixed up the
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	Therefore, $a_n = \pi r^2(2n-1)$ for $n=1, 2, 3, \dots, N$. The difference between any two consecutive terms (for $n=2,3,\dots,N$) $\begin{aligned} &= a_n - a_{n-1} \\ &= [\pi r^2(2n-1)] - \{\pi r^2[2(n-1)-1]\} \\ &= 2\pi r^2 \end{aligned}$ Since $2\pi r^2$ is independent of n , it is a constant. Since the difference between every two consecutive terms is constant, the sequence $a_1, a_2, a_3,\dots, a_N$ is an arithmetic progression. (shown)	A_i's (name of regions) with the a_i's (area of the regions), incorrectly labelling $\pi r^2(2n-1) = A_n - A_{n-1}$ as a constant. Checking the first few terms of a sequence share a common difference to conclude that the sequence forms an arithmetic progression is NOT allowed																																																												
(ii) [4]	The intensity of green colour in area $A_1, A_3, A_5, \dots, A_n, \dots, A_{N-1}$, where $N-1$ is an odd number, is: 1, $0.9, 0.9^2 \dots, 0.9^{\frac{N-1}{2}}$, which is a geometric progression $0.9^{\frac{N-2}{2}} < 0.25$ $\frac{N-2}{2} > \frac{\ln 0.25}{\ln 0.9}$ $N > 28.315 \quad (5\text{s.f.})$ Since N is an even integer, $N = 30$. Alternative 1 Since $\frac{N-2}{2}$ is an integer and $\frac{N-2}{2} > \frac{\ln 0.25}{\ln 0.9} = 13.158 \quad (5\text{s.f.})$, then $\frac{N-2}{2} = 14 \Rightarrow N = 30$. Alternative 2 <table border="1"><thead><tr><th>X</th><th>Y₁</th><th></th><th></th><th></th></tr></thead><tbody><tr><td>26</td><td>0.2824</td><td></td><td></td><td></td></tr><tr><td>27</td><td>0.2679</td><td></td><td></td><td></td></tr><tr><td>28</td><td>0.2542</td><td></td><td></td><td></td></tr><tr><td>29</td><td>0.2411</td><td></td><td></td><td></td></tr><tr><td>30</td><td>0.2288</td><td></td><td></td><td></td></tr><tr><td>31</td><td>0.217</td><td></td><td></td><td></td></tr><tr><td>32</td><td>0.2059</td><td></td><td></td><td></td></tr><tr><td>33</td><td>0.1953</td><td></td><td></td><td></td></tr><tr><td>34</td><td>0.1853</td><td></td><td></td><td></td></tr><tr><td>35</td><td>0.1758</td><td></td><td></td><td></td></tr><tr><td>36</td><td>0.1668</td><td></td><td></td><td></td></tr></tbody></table> X=26 From GC, (N is an even integer) when $N = 28$, $0.9^{\frac{N-2}{2}} = 0.2542 > 0.25$; when $N = 30$, $0.9^{\frac{N-2}{2}} = 0.2288 < 0.25$. $\therefore N = 30$	X	Y ₁				26	0.2824				27	0.2679				28	0.2542				29	0.2411				30	0.2288				31	0.217				32	0.2059				33	0.1953				34	0.1853				35	0.1758				36	0.1668				Candidate should take note of the alternative solutions. Two common mistakes committed were failing to flip the inequality sign when dividing by $\ln(0.9)$, a negative number, and forgetting that N is an even number, thus concluding that the least value of N is 29 instead of 30.
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Alternative 3

Let the intensity of the green-colour regions form a geometric progression of the form $x, 0.9x, 0.9^2x, \dots, 0.9^{n-1}x, \dots$ where the intensity for region n is given by $0.9^{n-1}x$. We want

$$0.9^{n-1}x < 0.25x$$

$$n-1 > \frac{\ln 0.25}{\ln 0.9}$$

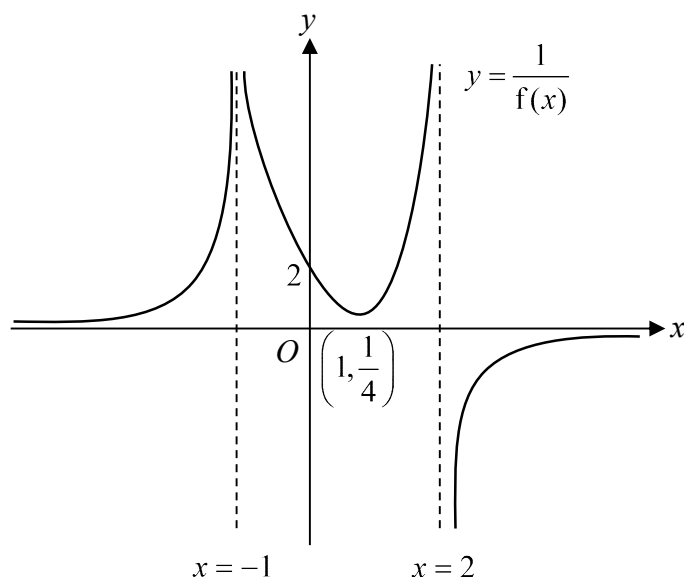
$$n > 14.158 \quad (5\text{s.f.})$$

The least value of n is 15. Hence the first region that satisfies the condition is the 15th odd region, A_{29} , and N is 30.

- 3** **(a)** Find $\int_0^1 \frac{9x - 4x^2}{9 - 4x^2} dx$, giving your answer in the form $p + q \ln 3 + r \ln 5$ where p , q and r are rational numbers to be determined. [5]
- (b)** Prove that $\sin 3x = 3 \sin x - 4 \sin^3 x$.
Hence, or otherwise, find $\int \sin x \sin 2x \sin 3x dx$. [4]

3(a) [5]	$\int_0^1 \frac{9x-4x^2}{9-4x^2} dx$ $= \int_0^1 \left(1 + \frac{9x-9}{9-4x^2} \right) dx$ $= 1+9\int_0^1 \frac{x}{9-4x^2} dx - 9\int_0^1 \frac{1}{9-4x^2} dx$ $= 1+9\left[-\frac{1}{8}\ln 9-4x^2 - \frac{1}{12}\ln\left \frac{3+2x}{3-2x}\right \right]_0^1$ $= 1+\frac{9}{8}\ln\frac{9}{5}-\frac{3}{4}\ln 5$ $= 1+\frac{9}{4}\ln 3-\frac{15}{8}\ln 5$	Pay attention to coefficients when integrating. 1st integral is of the form $\int \frac{f'(x)}{f(x)} dx = \ln f(x) + c$ 2nd integral is of the form $\int \frac{1}{a^2-x^2} dx = \frac{1}{2a} \ln \left \frac{a+x}{a-x} \right + c$ Check by differentiation.
3(b) [4]	$\sin 3x = \sin(2x+x)$ $= \sin 2x \cos x + \cos 2x \sin x$ $= 2 \sin x \cos^2 x + (1-2 \sin^2 x) \sin x$ $= 2 \sin x (1-\sin^2 x) + \sin x - 2 \sin^3 x$ $= 3 \sin x - 4 \sin^3 x$ $\int \sin x \sin 2x \sin 3x dx$ $= \int \sin x (2 \sin x \cos x)(3 \sin x - 4 \sin^3 x) dx$ $= 6 \int \sin^3 x \cos x dx - 8 \int \sin^5 x \cos x dx$ $= \frac{3}{2} \sin^4 x - \frac{4}{3} \sin^6 x + c$	Addition formula and double angle formula for sine and cosine. (MF 26) Don't forget to include the arbitrary constant c.

- 4 (a) The diagram below shows the curve of $y = \frac{1}{f(x)}$ where $f(x)$ is a polynomial. The curve has a minimum point at $\left(1, \frac{1}{4}\right)$ and cuts the y -axis at $(0, 2)$. The lines $x = -1$ and $x = 2$ are the vertical asymptotes and the line $y = 0$ is the horizontal asymptote to the curve.



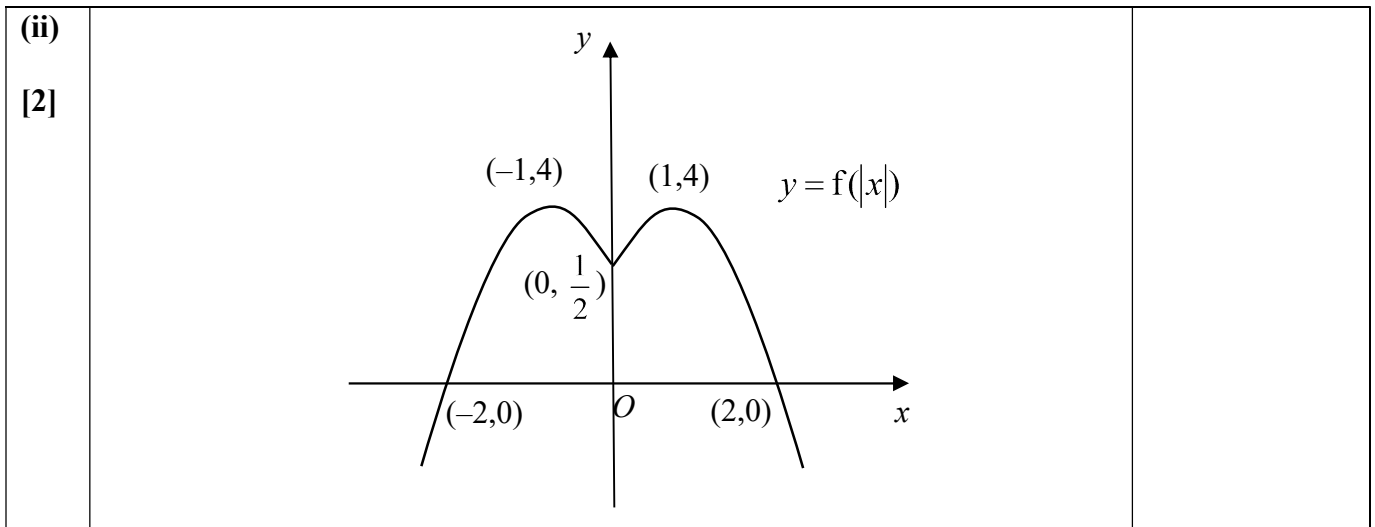
Sketch on separate diagrams, the graphs of

(i) $y = f(x)$,

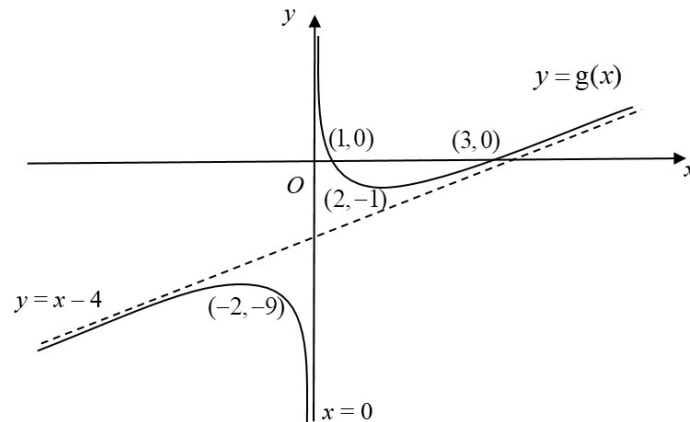
(ii) $y = f(|x|)$,

Labelling all relevant point(s).

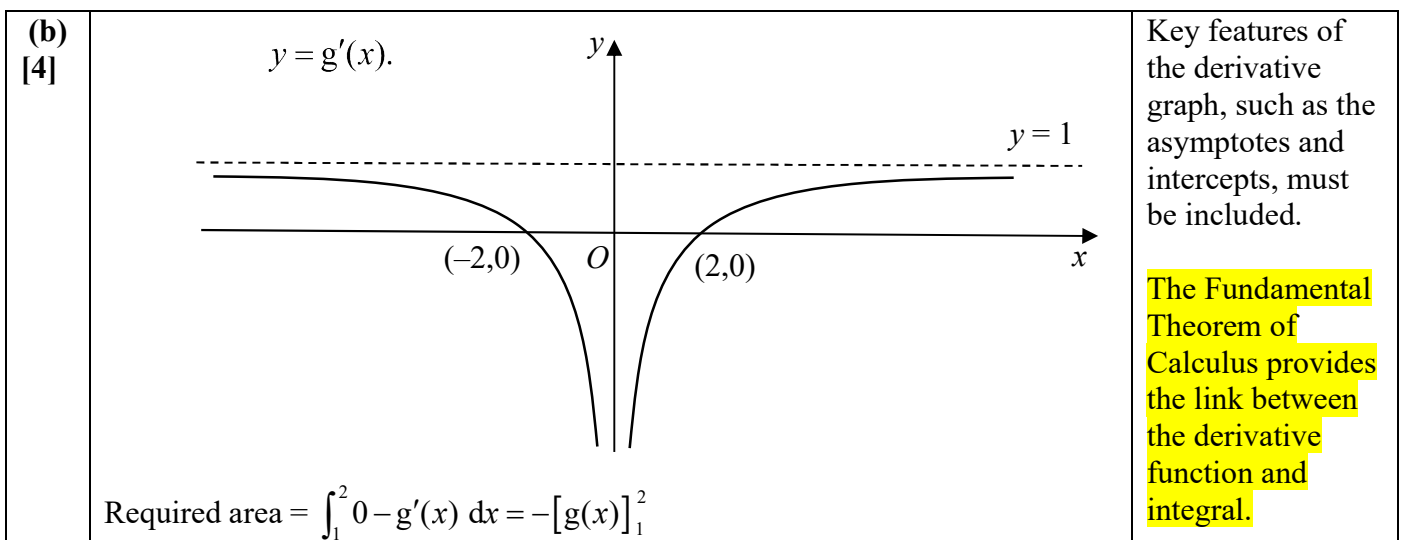
4a (i) [3]	The graph shows a smooth curve $y = f(x)$ on a Cartesian coordinate system. The curve passes through the points $(-1, 0)$, $(0, \frac{1}{2})$, $(1, 4)$, and $(2, 0)$. The origin is labeled O.	The fact that $f(x)$ is a polynomial means that the curve will be smooth. (no “holes” or “open circles” at the x-intercepts)
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- (b) The diagram below shows the curve of $y = g(x)$. The curve has a maximum point at $(-2, -9)$ and a minimum point at $(2, -1)$. The curve crosses the x -axis at $(1, 0)$ and $(3, 0)$. The line $x = 0$ is the vertical asymptote and the line $y = x - 4$ is the oblique asymptote to the curve.

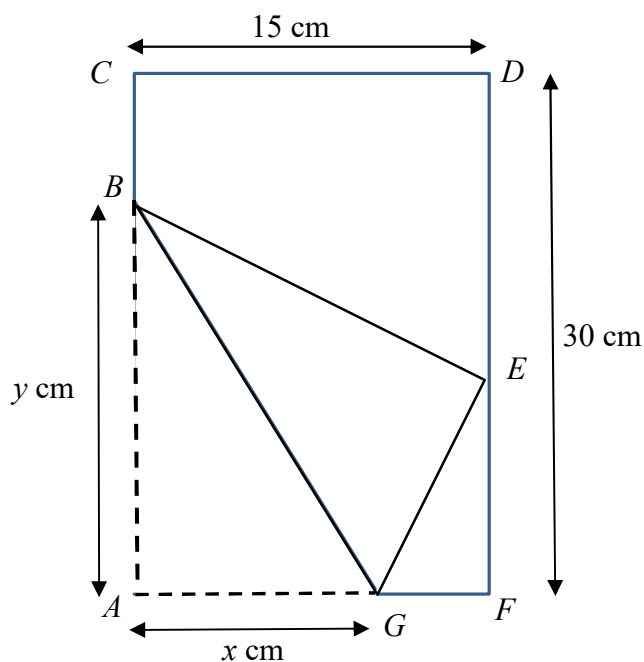


- (i) Sketch the graph of $y = g'(x)$, labelling all relevant point(s) and stating the equations of any asymptotes.
- (ii) Find the area bounded by the graph of $y = g'(x)$, the lines $x = 1$, $x = 2$ and the x -axis.
- [2]



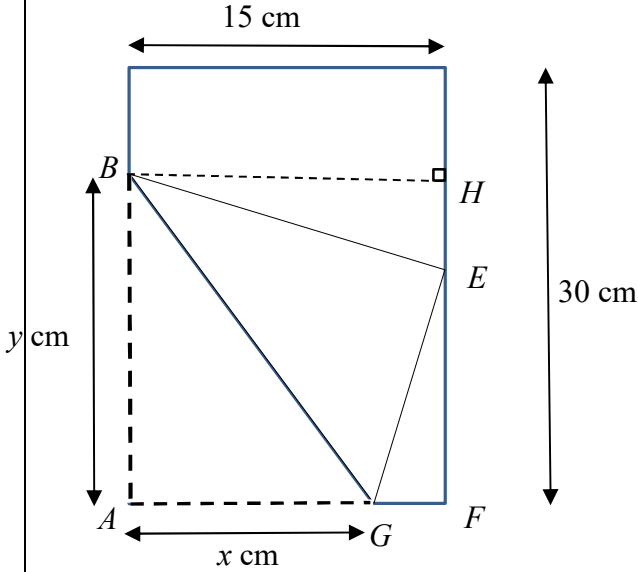
	$= -[g(2) - g(1)]$ $= -[-1 - 0] = 1$	Note the negative sign for the required area.
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5



Tom has a rectangular piece of paper $ACDF$ with length 30 cm and breadth 15 cm. He folds the lower left-hand corner, A , to reach the rightmost edge of the paper at E . After that he will cut out the triangle EFG and the trapezium $BCDE$ to obtain a kite shaped figure $ABEG$ with AB of length y cm and AG of length x cm.

- (i) Find the length of EF in terms of x . [2]
- (ii) Show that $y = \frac{x\sqrt{15}}{\sqrt{2x-15}}$. [2]
- (iii) Using differentiation, find the exact values of x and y which give the minimum area of the kite $ABEG$. [5]

5(i) [2]	Let the length of EF be w $w^2 = x^2 - (15 - x)^2$ $w^2 = x^2 - (15^2 - 30x - x^2)$ $\Rightarrow w^2 = 30x - 15^2$ $\Rightarrow w = \sqrt{15(2x - 15)}^{\frac{1}{2}}$	Pythagoras Theorem.
(ii) [2]	Method 1 (similar triangles)  Triangle BHE is similar to triangle EFG Therefore $\frac{BE}{EG} = \frac{BH}{EF} \Rightarrow \frac{y}{x} = \frac{15}{\sqrt{15(2x - 15)}^{\frac{1}{2}}}$ $\Rightarrow y = \frac{x\sqrt{15}}{\sqrt{(2x - 15)}}$ Method 2 (Pythagoras Theorem) $BH^2 + HE^2 = BE^2$ $15^2 + (y - EF)^2 = y^2$ $225 + (y - \sqrt{15}\sqrt{(2x - 15)})^2 = y^2$ $225 + y^2 - 2y\sqrt{15}\sqrt{(2x - 15)} + 15(2x - 15) = y^2$ $-2y\sqrt{15}\sqrt{(2x - 15)} + 30x = 0$ $y = \frac{x\sqrt{15}}{\sqrt{(2x - 15)}}$	Candidates to note the alternative methods. Note that this is a “show” question so working or justification “by similar triangles ...” needs to be explicitly stated.

(iii)

[5]

Area of kite $ABEG = xy$

Method 1 (find stationary point)

$$\therefore A = \frac{x^2 \sqrt{15}}{\sqrt{2x-15}}$$

$$\Rightarrow A^2 = \frac{15x^4}{2x-15}$$

$$\Rightarrow 2A \frac{dA}{dx} = \frac{60x^3(2x-15) - 2(15x^4)}{(2x-15)^2}$$

For stationary value, $\frac{dA}{dx} = 0$

$$\Rightarrow 30x^3 [2(2x-15) - x] = 0$$

$$\Rightarrow x = 0 \text{ (reject) or } x = 10$$

$$\text{When } x = 10, y = \frac{10\sqrt{15}}{\sqrt{5}} = 10\sqrt{3}$$

Method 2 (find stationary point)

$$\therefore A = \frac{x^2 \sqrt{15}}{\sqrt{2x-15}} = \sqrt{15}x^2 (2x-15)^{-\frac{1}{2}}$$

$$\therefore \frac{dA}{dx} = 2\sqrt{15}x(2x-15)^{-\frac{1}{2}} + \sqrt{15}x^2 \left[-\frac{1}{2}(2x-15)^{-\frac{3}{2}}(2) \right]$$

$$\Rightarrow \frac{dA}{dx} = x\sqrt{15}(2x-15)^{-\frac{3}{2}} [2(2x-15) - x]$$

$$\Rightarrow \frac{dA}{dx} = x\sqrt{15}(2x-15)^{-\frac{3}{2}} [3x-30]$$

For stationary value, $\frac{dA}{dx} = 0$

$$\Rightarrow x\sqrt{15}(2x-15)^{-\frac{3}{2}} [3x-30] = 0$$

$$\Rightarrow x = 0 \text{ (reject) or } x = 10$$

$$\text{When } x = 10, y = \frac{10\sqrt{15}}{\sqrt{5}} = 10\sqrt{3}$$

Method 1 (Determine nature of stationary point)

Using Second Derivative Test

$$\begin{aligned} \frac{d^2A}{dx^2} &= \sqrt{15} \left(-\frac{3}{2} \right) (2x-15)^{-\frac{5}{2}} [3x^2 - 30x] + \sqrt{15} (2x-15)^{-\frac{3}{2}} [6x-30] \\ &= 10.3923 > 0 \end{aligned}$$

Therefore $x = 10$, $y = 10\sqrt{3}$ gives the minimum area of the kite.

Method 2 (Determine nature of stationary point)

Using first derivative test

	$x = 10^-$	$x = 10$	$x = 10^+$
x	+ve	+ve	+ve
$3(x-10)$	-ve	0	+ve
$\frac{dA}{dx} = x\sqrt{15}(2x-15)^{-\frac{3}{2}}[3x-30]$	-ve	0	+ve

We conclude that $x = 10$, $y = 10\sqrt{3}$ gives the minimum area of the kite.

Marker Comments for Students

(iii) Candidate should read the question carefully as many erroneously minimized y instead of the area. Many students also have the **misconception** that since the question asked for a minimum area, they do not have to verify it. **Do note that for any max/min questions, showing that it is a min/max is a MUST.** 2nd Derivative test might be easier here.

For students who use the first derivative test, you need to identify the factor that resulted in the sign change. This table alone is not good enough.

	$x = 10^-$	$x = 10$	$x = 10^+$
$\frac{dA}{dx} = x\sqrt{15}(2x-15)^{-\frac{3}{2}}[3x-30]$	-ve	0	+ve

6 (a) It is given that $e^y = \tan\left(x + \frac{\pi}{3}\right)$.

(i) Show that $\frac{dy}{dx} = e^y + e^{-y}$. [2]

(ii) Using differentiation, find the Maclaurin series for $\ln\left[\tan\left(x + \frac{\pi}{3}\right)\right]$ up to and including the term in x^2 . [3]

(b) Let $f(x) = \frac{3x^2 - 4x + 5}{(1+x)(1-x)^2}$.

(i) Express $f(x)$ in the form $\frac{A}{(1+x)} + \frac{B}{(1-x)} + \frac{C}{(1-x)^2}$ where A , B and C are constants to be determined. [2]

(ii) Hence find the expansion of $f(x)$ up to and including the term in x^4 . [2]

(iii) Write down the coefficient of x^r in the expansion of $f(x)$ in terms of r . [1]

6a(i) [2]	$e^y = \tan\left(x + \frac{\pi}{3}\right)$ Differentiate with respect to x $e^y \frac{dy}{dx} = \sec^2\left(x + \frac{\pi}{3}\right)$ $e^y \frac{dy}{dx} = 1 + \tan^2\left(x + \frac{\pi}{3}\right)$ $e^y \frac{dy}{dx} = 1 + e^{2y}$ $\frac{dy}{dx} = e^y + e^{-y}$	Implicit differentiation/ chain rule $\frac{d}{dx} e^y = e^y \frac{dy}{dx}$ and $\frac{d}{dx} e^{-y} = -e^{-y} \frac{dy}{dx}$. Recall the trigo identity $\sec^2 \theta = \tan^2 \theta + 1$
(ii) [3]	$\frac{dy}{dx} = e^y + e^{-y}$ $\frac{d^2 y}{dx^2} = e^y \frac{dy}{dx} - e^{-y} \frac{dy}{dx}$ When $x = 0$, $e^y = \tan \frac{\pi}{3} \Rightarrow y = \ln(\sqrt{3}) = \frac{1}{2} \ln 3$ $\frac{dy}{dx} = e^{\frac{1}{2} \ln 3} + e^{-\frac{1}{2} \ln 3} \Rightarrow \frac{dy}{dx} = \sqrt{3} + \frac{1}{\sqrt{3}} = \frac{4}{\sqrt{3}}$ $\frac{d^2 y}{dx^2} = \frac{4}{\sqrt{3}} \left(\sqrt{3} - \frac{1}{\sqrt{3}} \right) = \frac{8}{3}$ $\ln \left[\tan \left(x + \frac{\pi}{3} \right) \right] = \frac{1}{2} \ln 3 + \frac{4}{\sqrt{3}} x + \frac{8}{2!} x^2 + \dots = \frac{1}{2} \ln 3 + \frac{4}{\sqrt{3}} x + \frac{4}{3} x^2$	Note the significance of finding the expression in part (i) here to find $\frac{d^2 y}{dx^2}$ Simplify all algebraic expressions.
b(i) [2]	$\frac{3x^2 - 4x + 5}{(1+x)(1-x)^2} = \frac{A}{(1+x)} + \frac{B}{(1-x)} + \frac{C}{(1-x)^2}$ $3x^2 - 4x + 5 = A(1-x)^2 + B(1+x)(1-x) + C(1+x)$ When $x = 1$, $3 - 4 + 5 = 2C \Rightarrow C = 2$ $x = -1$, $3 + 4 + 5 = 4A \Rightarrow A = 3$ $x = 0$, $5 = A + B + C \Rightarrow B = 0$	Candidates must show more care in getting the correct partial fractions.
(ii) [2]	$\frac{3}{(1+x)} + \frac{2}{(1-x)^2} = 3(1+x)^{-1} + 2(1-x)^{-2}$ $= 3(1 - x + x^2 - x^3 + x^4 + \dots)$ $+ 2 \left(1 + 2x + \frac{-2(-3)}{2!} x^2 + \frac{-2(-3)(-4)}{3!} (-x)^3 + \frac{-2(-3)(-4)(-5)}{4!} (-x)^4 + \dots \right)$ $= 3(1 - x + x^2 - x^3 + x^4 + \dots) + 2(1 + 2x + 3x^2 + 4x^3 + 5x^4 + \dots)$ $= 5 + x + 9x^2 + 5x^3 + 13x^4 + \dots$	Binomial series expansion from MF26.
(iii) [1]	The coefficient of x^r is $= 3(-1)^r + 2(r+1)$	Candidates to note general term.

7 Do not use a calculator in answering this question.

(a) Find the roots of the equation $w^2(1-i) + 4w + (10+10i) = 0$, giving your answers in cartesian form $a+ib$. [3]

(b) It is given that $z = -\sqrt{3} + i$.

(i) Find an exact expression for z^5 . Give your answer in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$. [2]

(ii) Find the three smallest positive whole number values of n for which $\frac{z^n}{iz^*}$ is purely imaginary. [4]

(iii) Given that $\left|1 + \frac{p}{z^*}\right| = \sqrt{7}$, find exactly the possible values of the real number p . [3]

7(a) [3]	$w^2(1-i) + 4w + (10+10i) = 0$ $w = \frac{-4 \pm \sqrt{4^2 - 4(1-i)(10+10i)}}{2(1-i)}$ $= \frac{-4 \pm \sqrt{16 - 40(1-i)(1+i)}}{2(1-i)}$ $= \frac{-4 \pm \sqrt{16 - 40(2)}}{2(1-i)}$ $= \frac{-4 \pm \sqrt{-64}}{2(1-i)}$ $= \frac{-4 \pm 8i}{2(1-i)} \times \frac{1+i}{1+i}$ $= (-1 \pm 2i)(1+i)$ $w = (-1+2i)(1+i) \quad \text{or} \quad w = (-1-2i)(1+i)$ $= -1-i+2i+2i^2 \quad = -1-i-2i-2i^2$ $= -3+i \quad = 1-3i$	Use quadratic formula. Recall Y6 CT2. The simplification of the intermediate answers $\frac{-4 \pm 8i}{2(1-i)}$ must be shown as the use of calculator is not allowed.
(b) (i) [2]	Given $z = -\sqrt{3} + i$, $z = 2$, $\arg(z) = \frac{5\pi}{6}$ Method 1 :	

$$|z^5| = 2^5 = 32$$

$$\arg(z^5) = 5 \arg(z) - 4\pi$$

$$= 5 \left(\frac{5\pi}{6} \right) - 4\pi$$

$$= \frac{25\pi}{6} - 4\pi$$

$$= \frac{\pi}{6}$$

$$\therefore z^5 = 32e^{i\frac{\pi}{6}}$$

Method 2 :

$$z^5 = \left(2e^{i\frac{5\pi}{6}} \right)^5$$

$$= 2^5 e^{i\left(\frac{25\pi}{6} - 4\pi\right)}$$

$$= 32e^{i\frac{\pi}{6}}$$

(ii)
[4]

$$\frac{z^n}{iz^*} = \frac{2^n e^{i\frac{5n\pi}{6}}}{e^{i\frac{\pi}{2}} 2e^{-i\frac{5\pi}{6}}}$$

$$= 2^{n-1} e^{i\left(\frac{5n\pi}{6} - \frac{\pi}{2} + \frac{5\pi}{6}\right)}$$

$$= 2^{n-1} e^{i\frac{(5n+2)\pi}{6}}$$

Alternatively : Just determine $\arg\left(\frac{z^n}{iz^*}\right)$

$$n \arg(z) - \arg(i) - \arg(z^*)$$

$$= n \left(\frac{5\pi}{6} \right) - \frac{\pi}{2} + \frac{5\pi}{6} \quad \text{since } \arg(z^*) = -\arg(z) = -\frac{5\pi}{6}$$

$$= \left(\frac{5n+2}{6} \right) \pi$$

For $\frac{z^n}{iz^*}$ to be purely imaginary,

$$\cos\left(\frac{5n+2}{6}\pi\right) = 0$$

$$\left(\frac{5n+2}{6}\right)\pi = (2k+1)\frac{\pi}{2}, \quad k \in \mathbb{Z}$$

$$5n+2 = 3(2k+1)$$

$$n = \frac{6k+1}{5}$$

$\therefore$ The three smallest positive whole number values of n are 5, 11, 17.

OR alternatively

Note that n is a positive integer.

	$\frac{z^n}{iz^*} = \frac{-iz^{n+1}}{ z ^2} = -2^{n-1}ie^{i\frac{5(n+1)\pi}{6}}$ For $\frac{z^n}{iz^*}$ to be purely imaginary, $\sin\left(\frac{5n+5}{6}\right)\pi = 0, \quad k \in \mathbb{Z}$ $\left(\frac{5n+5}{6}\right)\pi = k\pi, \quad k \in \mathbb{Z}$ $5n = 6k - 5$ $n = \frac{6k}{5} - 1$ $\therefore$ The three smallest positive whole number values of n are 5, 11, 17.	
(iii)	$\left 1 + \frac{p}{z^*}\right = \sqrt{7} \Leftrightarrow \left \frac{z^* + p}{z^*}\right = \sqrt{7}$ $\frac{ -\sqrt{3} - i + p }{ z^* } = \sqrt{7}$ $ (-\sqrt{3} + p) - i = 2\sqrt{7} \quad \text{since } z^* = z = 2$ $(-\sqrt{3} + p)^2 + (-1)^2 = (2\sqrt{7})^2$ $(-\sqrt{3} + p)^2 = 27$ $-\sqrt{3} + p = \pm\sqrt{27}$ $p = \sqrt{3} \pm 3\sqrt{3}$ $\therefore p = 4\sqrt{3} \text{ or } -2\sqrt{3}$	Simplify algebraic expressions
	Marker Comments for Students (iii) Candidates should understand why (a) $\left 1 + \frac{p}{z^*}\right \neq 1 + \left \frac{p}{z^*}\right$ (b) $\left 1 + \frac{p}{z^*}\right ^2 \neq \left(1 + \frac{p}{z^*}\right)^2$ (c) $\left 1 + \frac{p}{z^*}\right = \sqrt{7}$ does not imply $1 + \frac{p}{z^*} = \pm\sqrt{7}$	

- 8 The function $f(x) = a - (1+x)^{\frac{1}{3}}$ is defined for $0 \leq x \leq 7$, where a is a positive real constant.

(i) Solve the inequality $f(x) \geq a - 1$. [2]

(ii) Show that the graph of $y = f(x)$ has a negative gradient at all points on the graph. Find the range of f . [2]

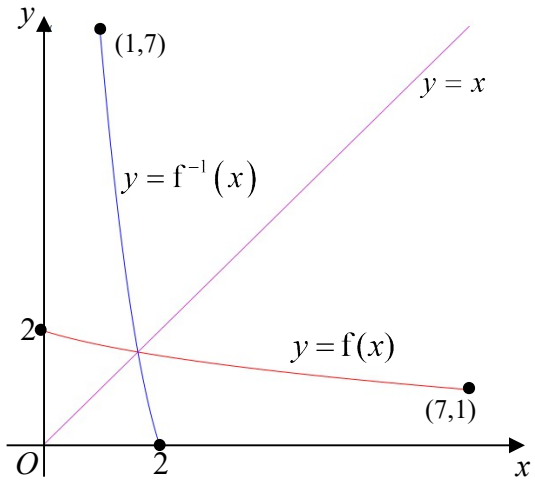
Use $a = 3$ in the rest of the question.

(iii) Find $f^{-1}(x)$. [2]

(iv) Sketch on the same diagram the graphs of $y = f(x)$ and $y = f^{-1}(x)$, showing clearly the geometrical relationship between the two graphs and the line $y = x$. [3]

(v) Show that the x -coordinate of the point of intersection of the graphs of $y = f(x)$ and $y = f^{-1}(x)$ satisfies the equation $x^3 - 9x^2 + 28x - 26 = 0$.
Hence find the solution of the equation $f(x) = f^{-1}(x)$. [3]

8(i) [2]	$a - (1+x)^{\frac{1}{3}} \geq a - 1$ $(1+x)^{\frac{1}{3}} \leq 1$ $x \leq 0$ But $0 \leq x \leq 7$, Solution is $x = 0$	Note the restriction on solutions for x because of the domain of f .
(ii) [2]	$f'(x) = -\frac{1}{3}(1+x)^{-\frac{2}{3}} = -\frac{1}{3} \left(\frac{1}{\sqrt[3]{1+x}} \right)^2 < 0$ Thus, the graph of $y = f(x)$ has a negative gradient at all points on the graph Since f is continuous and decreasing over $0 \leq x \leq 7$, $f(7) \leq f(x) \leq f(0)$ $a - 2 \leq f(x) \leq a - 1$ $\therefore R_f = [a - 2, a - 1]$	
(iii) [2]	$y = 3 - (1+x)^{\frac{1}{3}}$ $1+x = (3-y)^3$ $x = (3-y)^3 - 1$ $f^{-1}(x) = (3-x)^3 - 1$	Make x the subject

(iv) [3]		Use the same scale along the x and y axis. Note the domain of f and indicate the end points. Draw a sketch so that the curves appear as reflections about $y=x$.
(v) [3]	From the graphs of $y = f(x)$ and $y = f^{-1}(x)$, we see that $f(x) = f^{-1}(x)$ exactly where $f^{-1}(x) = x$. $(3-x)^3 - 1 = x$ $27 - 27x + 9x^2 - x^3 - 1 = x$ $x^3 - 9x^2 + 28x - 26 = 0 \text{ (shown)}$ From GC, $x = 1.62$ (3sf)	Do not include complex solutions!

- 9 (a) The variables x and y are related by the differential equation $\frac{dy}{dx} = \sqrt{x}e^{\sqrt{x}}$.

Using the substitution $u = \sqrt{x}$, show that $y = k \int u^2 e^u du$ where k is a constant to be determined. Hence find the general solution of the differential equation. [5]

- (b) Ms Frugal bought a second-hand oven which has a broken timer. Each time she bakes she uses the timer on her handphone. One day, she decides to make a loaf of bread. She takes the fermented dough out of her refrigerator and checks that the internal temperature of the dough is 4°C . She puts it into the oven which has been pre-heated to a constant temperature of 180°C and forgets to set the timer on her handphone. She also does not note the time. At 10.06 am, she realizes that the timer has not been set. She checks her dough which now has an internal temperature of 80°C . Twenty minutes later, she checks the dough again and the internal temperature has risen to 130°C .

Newton's Law of Cooling states that the rate of increase of the temperature $\theta^\circ\text{C}$ of an object after t minutes is directly proportional to the difference in the temperatures of the object and its surrounding.

The dough has to be baked at a constant oven temperature of 180°C for one and a half hour to cook through. By forming a differential equation, find the time (to the nearest minute) that Ms Frugal should remove the dough from the oven.

[8]

9(a) [5]	$u = \sqrt{x} \Rightarrow \frac{du}{dx} = \frac{1}{2\sqrt{x}} \Rightarrow dx = 2u du$ $\frac{dy}{dx} = \sqrt{x}e^{\sqrt{x}}$ $y = \int \sqrt{x}e^{\sqrt{x}} dx$ $= \int ue^u \cdot 2u du$ $= 2 \int u^2 e^u du$ $= 2 \left(u^2 e^u - 2 \int ue^u du \right)$ $= 2u^2 e^u - 4 \left(ue^u - \int e^u du \right)$ $= 2u^2 e^u - 4ue^u + 4e^u + c$ $= 2xe^{\sqrt{x}} - 4\sqrt{x}e^{\sqrt{x}} + 4e^{\sqrt{x}} + c$	Substitution technique for DE Integration by parts Final expression for y in terms of x
9(b) [8]	$\frac{d\theta}{dt} = k(180 - \theta) \text{ where } k \text{ is positive}$ $\frac{1}{180 - \theta} \frac{d\theta}{dt} = k$ Integrate w.r.t t, $\int \frac{1}{180 - \theta} d\theta = \int k dt$ $-\ln(180 - \theta) = kt + c, \text{ where } \theta < 180$ When $t = 0, \theta = 4$, $c = -\ln 176$ $\therefore -\ln(180 - \theta) = kt - \ln 176$ Let t_0 be the time elapsed from 4°C to 80°C. When $t = t_0, \theta = 80$, $-\ln 100 = kt_0 - \ln 176 \Rightarrow kt_0 = \ln \frac{44}{25} \quad \dots(1)$ When $t = t_0 + 20, \theta = 130$, $-\ln 50 = k(t_0 + 20) - \ln 176 \Rightarrow k(t_0 + 20) = \ln \frac{88}{25} \quad \dots(2)$ By (1) and (2), $\frac{t_0}{t_0 + 20} = \frac{\ln \frac{44}{25}}{\ln \frac{88}{25}}$ $t_0 = 16.312 \text{ (5s.f.)}$ Remaining number of minutes in oven after 10.06am $= 90 - 16.312$ $= 73.688 \text{ mins}$ Thus the time that Ms Frugal should take out the dough is 11.20am.	Minus sign or $+c$ being omitted in $-\ln(180 - \theta) = kt + c$ should be avoided. Absolute value for $180 - \theta$ is not necessary since $180 - \theta > 0$ from the context of the question. The time elapsed between 4°C and 130°C should be $t_0 + 20$ where t_0 is the time taken for the temperature to increase from 4°C to 80°C. The time required can be obtained without making θ the subject of the formula.

Alternative

$$\frac{d\theta}{dt} = k(180 - \theta) \text{ where } k \text{ is positive}$$

$$\frac{1}{180 - \theta} \frac{d\theta}{dt} = k$$

Integrate w.r.t t ,

$$\int \frac{1}{180 - \theta} d\theta = \int k dt$$

$$-\ln(180 - \theta) = kt + c, \text{ where } \theta < 180$$

When $t = 0, \theta = 80$,

$$c = -\ln 100$$

$$\therefore -\ln(180 - \theta) = kt - \ln 100$$

When $t = 20, \theta = 130$,

$$-\ln 50 = k(20) - \ln 100$$

$$k = \frac{\ln 2}{20}$$

$$\therefore -\ln(180 - \theta) = \left(\frac{\ln 2}{20}\right)t - \ln 100$$

When $\theta = 4$,

$$-\ln 176 = \left(\frac{\ln 2}{20}\right)t - \ln 100$$

$$t = -16.312$$

Remaining number of minutes in oven after 10.06am

$$= 90 - 16.312$$

$$= 73.688 \text{ mins}$$

Thus the time that Ms Frugal should take out the dough is 11.20am.

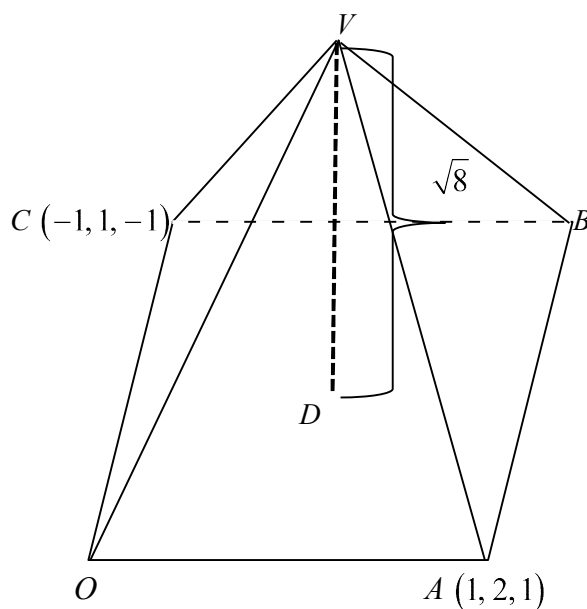
- 10** Two children Hansel and Gretel are participating in the junior category of the Festival of Lights competition. They decided to set up a structure consisting of a pyramid with a rectangular base $OABC$ and vertex V . The main power supply switch is positioned at O with coordinates $(0, 0, 0)$ and relative to O , the coordinates of A and C are $(1, 2, 1)$ and $(-1, 1, -1)$ respectively. The vertex V is fixed at a height $\sqrt{8}$ metres above the rectangular base $OABC$ such that it is equidistant to the points O, A, B and C . All the surfaces of the pyramid, excluding the base $OABC$, are completely covered with LED (light emitting diodes) light strips so that it illuminates in the dark.

- (i) Point D is the foot of the perpendicular from V to the rectangular base $OABC$. Show that the coordinates of D is $(0, 1.5, 0)$. [1]
- (ii) Find a vector perpendicular to the rectangular base $OABC$. Hence, find the position vector of V , given that its $\mathbf{k}$ -component is positive. [4]

Hansel and Gretel decide to install two different colour display schemes on the pyramid and the control switch is to be installed inside the pyramid at a point E with position vector $\alpha\mathbf{i} + 1.5\mathbf{j} - \alpha\mathbf{k}$, where α is a constant.

- (iii) Show that VE is perpendicular to the rectangular base $OABC$ and explain why $-2 < \alpha < 0$. [3]
- (iv) Find the distance between E and the surface OVA in terms of α , simplifying your answer. [4]
- (v) Given that the ratio of the distance between E and the rectangular base $OABC$ to the distance between E and the surface OVA is $10:\sqrt{105}$, find the value of α . [2]

10(i)
[1]



Since V is equidistant to the points O, A, B and C , and point D is the foot of perpendicular from V to the rectangular base $OABC$, by symmetry D is the centre of the rectangular $OABC$. In other

	words, D is the midpoint of AC. $\overrightarrow{OD} = \frac{1}{2}(\overrightarrow{OA} + \overrightarrow{OC}) = \frac{1}{2} \left[\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix} \right] = \begin{pmatrix} 0 \\ 1.5 \\ 0 \end{pmatrix}$ $\therefore$ Co-ordinates of D is $(0, 1.5, 0)$. (shown)	Use the Ratio Theorem to find the midpoint easily.
(ii) [4]	A vector perpendicular to $OABC$ $= \overrightarrow{OA} \times \overrightarrow{OC} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \times \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} -3 \\ 0 \\ 3 \end{pmatrix}$ Since the $\mathbf{k}$-component of V is positive, $\overrightarrow{DV} = \frac{\sqrt{8}}{\sqrt{2}} \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ 0 \\ 2 \end{pmatrix}$ $\overrightarrow{DV} = \overrightarrow{OV} - \overrightarrow{OD}$ $\overrightarrow{OV} = \begin{pmatrix} -2 \\ 0 \\ 2 \end{pmatrix} + \begin{pmatrix} 0 \\ 1.5 \\ 0 \end{pmatrix} = \begin{pmatrix} -2 \\ 1.5 \\ 2 \end{pmatrix}$	Cross 2 vectors parallel to the plane $OABC$ which are <i>non-parallel</i> to each other. Easiest to use the given vectors $\overrightarrow{OA}$ and $\overrightarrow{OC}$. Common mistakes include $\overrightarrow{VD} = \begin{pmatrix} 0 \\ 1.5 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix},$ $ \overrightarrow{OV} = \sqrt{8} \text{ and } \overrightarrow{OV} = \begin{pmatrix} x \\ y \\ \sqrt{8} \end{pmatrix}.$
(iii) [2]	Method 1: $\overrightarrow{VE} = \overrightarrow{OE} - \overrightarrow{OV} = \begin{pmatrix} \alpha \\ 1.5 \\ -\alpha \end{pmatrix} - \begin{pmatrix} -2 \\ 1.5 \\ 2 \end{pmatrix} = \begin{pmatrix} \alpha + 2 \\ 0 \\ -\alpha - 2 \end{pmatrix} = (\alpha + 2) \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$ Since $(\alpha + 2)$ is a constant, $\overrightarrow{VE}$ is parallel to $\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$, a vector normal to the plane $OABC$. Thus VE is perpendicular to the rectangular base $OABC$ (shown). Method 2: $\overrightarrow{VE} \cdot \overrightarrow{OA} = (\alpha + 2) \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = (\alpha + 2)(1 - 1) = 0$ $\overrightarrow{VE} \cdot \overrightarrow{OC} = (\alpha + 2) \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix} = (\alpha + 2)(-1 + 1) = 0$ Since $\overrightarrow{VE}$ is perpendicular to $\overrightarrow{OA}$ and $\overrightarrow{OC}$, which are two non-parallel vectors parallel to plane $OABC$, VE is perpendicular to the rectangular base $OABC$ (shown).	To show 2 vectors are parallel, express one as a multiple of the other. [Incorrect to write $\overrightarrow{VE} = \begin{pmatrix} 2 \\ 0 \\ -2 \end{pmatrix} + \alpha \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \text{ and}$ conclude that the “direction vector” of $\overrightarrow{VE}$ is parallel to $\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$]. Alternatively, perform 2 dot products. Insufficient to do only 1 dot product, e.g. $\begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} \perp \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$ but $\begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}$ is not perpendicular to plane $OABC$.

	Method 1: $\overrightarrow{VE} = (\alpha + 2) \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} = \left(\frac{\alpha + 2}{2} \right) \overrightarrow{VD}$ Since E is between V & D, $0 < \frac{\alpha + 2}{2} < 1 \Rightarrow -2 < \alpha < 0$. Method 2: Equation of line VD: $\mathbf{r} = \begin{pmatrix} 0 \\ 1.5 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$, $\lambda \in \mathbb{R}$ We have $\overrightarrow{OV}$ and $\overrightarrow{OD}$ with $\lambda = -2$ and $\lambda = 0$ respectively. $\overrightarrow{OE} = \begin{pmatrix} \alpha \\ 1.5 \\ -\alpha \end{pmatrix} = \begin{pmatrix} 0 \\ 1.5 \\ 0 \end{pmatrix} + \alpha \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \text{ satisfies the equation of line } VD$ with $\lambda = \alpha$. Since E is between V and D, $-2 < \alpha < 0$.	Or else show $\overrightarrow{VE} \times \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$
(iv) [4]	Normal of plane $OVA = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \times \begin{pmatrix} -2 \\ 3/2 \\ 2 \end{pmatrix} = \begin{pmatrix} 2.5 \\ -4 \\ 5.5 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 5 \\ -8 \\ 11 \end{pmatrix}$ $\text{Distance of } E \text{ to plane } OVA = \frac{\left \overrightarrow{OE} \cdot \begin{pmatrix} 5 \\ -8 \\ 11 \end{pmatrix} \right }{\left \begin{pmatrix} 5 \\ -8 \\ 11 \end{pmatrix} \right } = \frac{\left \begin{pmatrix} \alpha \\ 1.5 \\ -\alpha \end{pmatrix} \cdot \begin{pmatrix} 5 \\ -8 \\ 11 \end{pmatrix} \right }{\sqrt{210}}$ $= \frac{ -6\alpha - 12 }{\sqrt{210}}$ $= \frac{6\alpha + 12}{\sqrt{210}} \text{ since } -2 < \alpha < 0$	Be clear when to use dot product or cross product. Distance between a point and a plane can be found easily using $\frac{ \overrightarrow{VE} \cdot \overrightarrow{EF} }{ \overrightarrow{EF} }$. Other methods such as $\left[\begin{pmatrix} \alpha \\ 1.5 \\ -\alpha \end{pmatrix} + \mu \begin{pmatrix} 5 \\ -8 \\ 11 \end{pmatrix} \right] \cdot \begin{pmatrix} 5 \\ -8 \\ 11 \end{pmatrix} = 0$ or $\frac{ \overrightarrow{VE} \times \overrightarrow{VM} }{ \overrightarrow{VM} }$ where $\overrightarrow{OM} = \frac{1}{2} \overrightarrow{OA}$, work too.
(v) [2]	Distance between E and $OABC$ $= \overrightarrow{ED} = \left \alpha \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right = -\alpha\sqrt{2} \text{ (since } \alpha < 0)$ Given $\sqrt{105}(-\alpha\sqrt{2}) = 10 \left(\frac{6\alpha + 12}{\sqrt{210}} \right)$ $\Rightarrow -210\alpha = 60\alpha + 120 \Rightarrow \alpha = -\frac{4}{9}$	Need to consider that $-2 < \alpha < 0$, so $\overrightarrow{DE} = \sqrt{2\alpha^2} = -\alpha\sqrt{2}$.
	Marker Comments for Students Students should manage their time effectively to ensure sufficient time for a proper attempt for the question. Good to see many candidates drawing a diagram of the pyramid. Candidate should note the various methods for each part.	